Lecture 5. Many particle states

* Generalized Fourier transform

Consider

$$S(\mathbb{R}) := \left\{ \phi \in C^\infty(\mathbb{R}, \mathbb{C}) \text{ s.t. } \sup_{x \in \mathbb{R}} \left| x^\alpha \frac{\partial \phi}{\partial x^\beta} \right| < \infty \ \forall \alpha, \beta \geq 0 \right\}$$

(Schwartz space)

→ $D(\mathbb{R}) \underset{\text{dense}}{\subset} S(\mathbb{R})$

→ $S'(\mathbb{R})$: space of tempered distribution.

It is widely known that Fourier transform F,

$F : S(\mathbb{R}) \to S(\mathbb{R})$ is an automorphism.

If $T \in S'(\mathbb{R})$, then we define $FT \in S'(\mathbb{R})$ as

$$(FT)(\phi) := T(F\phi) \quad \forall \phi \in S(\mathbb{R})$$

→ Fourier transform of (Tempered distribution) $\psi(x) = e^{iP_0 x}$ is:

$$(F\psi)(\phi) = \psi(F\phi) \quad \forall \phi \in S(\mathbb{R})$$

$$\left(:= \int_{\mathbb{R}} e^{iP_0 x} \int_{\mathbb{R}} \phi(y) e^{-ixy} \, dy \, dx \right)$$

$$\int_{\mathbb{R}} e^{iP_0 x} \int_{\mathbb{R}} \phi(y) e^{-ixy} \, dy \, dx \overset{\text{DCT}}{=} \lim_{\varepsilon \to 0} \int_{\mathbb{R}} e^{iP_0 x} \cdot e^{-\varepsilon x^2} \int_{\mathbb{R}} \phi(y) e^{-ixy} \, dy \, dx$$

$$= \lim_{\varepsilon \to 0} \int_{\mathbb{R}} \int_{\mathbb{R}} \underbrace{\phi(y) e^{ix(P_0 - y) - \varepsilon x^2}}_{\text{abs integrable}} \, dy \, dx$$

$$\overset{\text{Fubini}}{=} \lim_{\varepsilon \to 0} \int_{\mathbb{R}} \int_{\mathbb{R}} \quad \text{''} \quad dx \, dy.$$

$$= \lim_{\varepsilon \to 0} \int_{\mathbb{R}} \phi(y) \sqrt{\frac{\pi}{\varepsilon}} \, e^{-\frac{(P_0 - y)^2}{4\varepsilon}} \, dy$$

$$\underset{y = 2\sqrt{\varepsilon} z + P_0}{=} \lim_{\varepsilon \to 0} \int_{\mathbb{R}} \phi(2\sqrt{\varepsilon} z + P_0) \, 2\sqrt{\pi} \, e^{-z^2} \, dz = 2\pi \phi(P_0)$$

$$\to (F\psi)(\phi) = 2\pi \delta_{P_0}(\phi) \to \hat{\psi} = 2\pi \delta_{P_0}$$

Lecture 5. Many particle states

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$$\underset{y = 2\sqrt{\varepsilon} z + p_0}{=} \lim_{\varepsilon \to 0} \int_{\mathbb{R}} \phi(2\sqrt{\varepsilon} z + p_0) \, 2\sqrt{\pi} \, e^{-z^2} \, dz = 2\pi \phi(p_0)$$

$$\rightarrow (F\psi)(\phi) = 2\pi \delta_{p_0}(\phi) \rightarrow \hat{\psi} = 2\pi \delta_{p_0}$$

$H_x = -\frac{1}{2m}\frac{d^2}{dx^2}$, $\psi_t = e^{-itH_x}\psi$. $\varphi = \hat{\psi}$. $\varphi_t = e^{-\frac{it}{2m}p^2}\varphi(p)$. $H_p = \frac{p^2}{2m}$

$$\rightarrow \quad \widehat{-\frac{1}{2m}\frac{d^2}{dx^2}\psi} = \frac{p^2}{2m}\hat{\psi}$$

(Basic property of Fourier transform)
$\widehat{\psi'} = ip\hat{\psi}$.

5.2. Schrödinger & Heisenberg pictures

state	$\psi_t = e^{-itH}\psi$	$\psi_t = \psi$.
observable operator	$A_t = A$.	$A_t = e^{itH}Ae^{-itH}$.
expected value.	$\langle\psi\vert e^{itH}Ae^{-itH}\vert\psi\rangle$	$\langle\psi\vert e^{itH}Ae^{-itH}\vert\psi\rangle$

equivalence.

- Tensor product state spaces

$$H^{\otimes n} := \overline{\operatorname{span}\{\psi_1 \otimes \cdots \otimes \psi_n,\ \psi_i \in H\}}$$

Where: $\psi_1 \otimes \cdots \otimes (\alpha\psi_i) \otimes \cdots \otimes \psi_n = \alpha(\psi_1 \otimes \cdots \otimes \psi_n)$ (scalar multiples)

$\psi_1 \otimes \cdots \otimes (\psi_i + \psi_i') \otimes \cdots \otimes \psi_n = (\psi_1 \otimes \cdots \otimes \psi_n) + (\psi_1 \otimes \cdots \otimes \psi_i' \otimes \cdots \otimes \psi_n)$
(distributivity)

$$(\psi_1 \otimes \cdots \otimes \psi_n, \phi_1 \otimes \cdots \otimes \phi_n) := (\psi_1, \phi_1)\cdots(\psi_n, \phi_n)$$

If $H = L^2(\mathbb{R})$, then $H^{\otimes n} \cong L^2(\mathbb{R}^n)$

$(\psi_1 \otimes \cdots \otimes \psi_n)(x_1, \ldots, x_n) = \psi_1(x_1)\cdots\psi_n(x_n) \quad \forall\, \psi_i \in L^2(\mathbb{R}),\ x_i \in \mathbb{R}$

Lecture 6. Bosonic Fock space

There are two kind of elementary particles in standard model:

Boson (Gluon, W/Z boson, photon, Higgs boson, etc) Bose-Einstein statistics

Fermion (quark, Lepton, etc) Fermi-Dirac statistics

: n identical particles are indistinguishable.

Let $\widehat{P_{ij}}$: exchange operator s.t.

$$\widehat{P_{ij}}\,\Phi(x_1,\dots,x_i,\dots,x_j,\dots,x_n) = \Phi(x_1,\dots,x_j,\dots,x_i,\dots,x_n)$$

Clearly, $\widehat{P_{ij}}$ is selfadjoint and $\widehat{P_{ij}}^2 = Id$.

→ eigenvalues are either 1 or −1

→ eigenspaces E_1^{ij}, E_{-1}^{ij}

We set $\mathcal{H}_{Sym}^{\otimes n} := \bigcap_{i\neq j} E_1^{ij}$, $\mathcal{H}_{Alt}^{\otimes n} := \bigcap_{i\neq j} E_{-1}^{ij}$

$$\rightarrow \mathcal{H}_{Sym}^{\otimes n} = \overline{\mathrm{span}\left\{\sum_{\sigma\in S_n} \psi_{\sigma(1)}\otimes\cdots\otimes\psi_{\sigma(n)} : \psi_1,\dots,\psi_n\in\mathcal{H}\right\}}$$

$$\mathcal{H}_{Alt}^{\otimes n} = \overline{\mathrm{span}\left\{\sum_{\sigma\in S_n} (-1)^{|\sigma|}\psi_{\sigma(1)}\otimes\cdots\otimes\psi_{\sigma(n)} : \psi_1,\dots,\psi_n\in\mathcal{H}\right\}}$$

Main postulate : state of n identical Bosons $\in \mathcal{H}_{Sym}^{\otimes n}$

" " " Fermions $\in \mathcal{H}_{Alt}^{\otimes n}$.

↳ If $\psi_i = \psi_j$ for some $i\neq j$, then

$$\sum_{\sigma\in S_n} (-1)^{|\sigma|}\psi_{\sigma(1)}\otimes\cdots\otimes\psi_{\sigma(n)} = 0.$$

→ $\mathcal{H}_{Alt}^{\otimes n}$ does not have such Building blocks

(i.e. Pauli exclusion principle)

- Time evolution on $\mathcal{H}^{\otimes n}$.

If $(U(t))_{t\in\mathbb{R}}$ is a unitary group, then

$$U(t)(\psi_1\otimes\cdots\otimes\psi_n) := (U(t)\psi_1)\otimes\cdots\otimes(U(t)\psi_n)$$

The corresponding self-adjoint operator $A(=-H)$ is defined as

$$A x = \lim_{t\to 0}\frac{U(t)x - x}{it}$$ by Stone's thm.

$$\therefore H(\psi_1\otimes\cdots\otimes\psi_n) = -\lim_{t\to 0}\frac{1}{it}\left(U(t)(\psi_1\otimes\cdots\otimes\psi_n) - \psi_1\otimes\cdots\otimes\psi_n\right)$$

$$= -\lim_{t\to 0}\frac{1}{it}\sum_{j=1}^{n} U(t)\psi_1\otimes\cdots\otimes U(t)\psi_{j-1}\otimes(U(t)\psi_j - \psi_j)\otimes\psi_{j+1}\otimes\cdots\otimes\psi_n$$

$$= \sum_{j=1}^{n}\psi_1\otimes\cdots\otimes\psi_{j-1}\otimes H\psi_j\otimes\psi_{j+1}\otimes\cdots\otimes\psi_n$$

- In particular, if $\mathcal{H} = L^2(\mathbb{R})$, $H = -\frac{1}{2m}\frac{d^2}{dx^2}$.

$\to \mathcal{H}^{\otimes n} = L^2(\mathbb{R}^n)$,

$$H(\psi_1\otimes\cdots\otimes\psi_n) = \sum_{j=1}^{n}\psi_1\otimes\cdots\otimes\psi_{j-1}\otimes\left(-\frac{1}{2m}\psi_j''\right)\otimes\psi_{j+1}\otimes\cdots\otimes\psi_n$$

$$H(\psi_1\otimes\cdots\otimes\psi_n)(x_1,\cdots,x_n) = -\frac{1}{2m}\sum_{j=1}^{n}\frac{\partial^2}{\partial x_j^2}\left(\psi_1(x_1)\cdots\psi_n(x_n)\right)$$

$$= -\frac{1}{2m}\Delta(\psi_1\otimes\cdots\otimes\psi_n)$$

$\to$ $H\psi = -\frac{1}{2m}\Delta\psi$ $\forall\psi\in D(\Delta)$: domain of the unique self-adj extension of Δ.

• Bosonic Fock space

$\mathcal{B} = \{(\psi_0, \psi_1, \cdots) : \psi_n \in \mathcal{H}_{sym}^{\otimes n}, \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty\}$

inner product of $\psi = (\psi_0, \psi_1, \cdots)$ and $\phi = (\phi_0, \phi_1, \cdots)$ is

$(\psi, \phi) = \sum_{i=0}^{\infty} (\psi_n, \phi_n)$ → Hilbert space

system has n particles with prob $\frac{\|\psi_n\|^2}{\|\psi\|^2}$.

$\mathcal{B}_0 := \bigoplus_{n=0}^{\infty} \mathcal{H}_{sym}^{\otimes n} \underset{\text{dense.}}{\subset} \mathcal{B}$

we can do similar work to Fermion:

$\mathcal{F} := \{(\psi_0, \psi_1, \cdots) : \psi_n \in \mathcal{H}_{Alt}^{\otimes n}, \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty\}$

inner product also defined as $(\psi, \phi) = \sum_{n=0}^{\infty} (\psi_n, \phi_n)$

$\mathcal{F}_0 := \bigoplus_{n=0}^{\infty} \mathcal{H}_{Alt}^{\otimes n} \underset{\text{dense}}{\subset} \mathcal{F}$

This process constructing Fock space, counting the number of particles in a given state in occupation number basis $|n_1, n_2, \cdots\rangle$, is called a second quantization

We can also use $|n_1, n_2, \cdots\rangle$ to Fermion, where

$|n_1, n_2 \cdots\rangle =$ (normalization of) $\sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}} \times (-1)^{|\sigma|}$

where n_i: number of i's in $\{m_1, \cdots, m_n\}$

→ $n_i = 0$ or 1. $\forall i$. (Pauli's principle)

If $\{e_1, e_2, \cdots\}$ is an ONB of $\mathcal{H}$, then

$\left\{\sum_{\sigma \in S(n)} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}}, \quad m_1, \cdots, m_n \in \mathbb{N}\right\}$ is an orthogonal basis of $\mathcal{H}_{Sym}^{\otimes}$

inner product with $\sum_{\sigma \in S(n)} e_{m'_{\sigma(1)}} \otimes \cdots \otimes e_{m'_{\sigma(n)}}$ is nonzero only if

$\{m_1, \cdots, m_n\} = \{m_1', \cdots, m_n'\}$.

i.e. $n_i = |\{j : m_j = i, 1 \le j \le n\}|$
$n_i' = |\{j = m_j' = i, 1 \le j \le n\}|$ are equal for every i.

$$\left\| \sum_{\sigma \in S(n)} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}} \right\|^2$$

$$= \sum_{\sigma, \tau \in S(n)} \delta_{m_{\sigma(1)} m_{\tau(1)}} \cdots \delta_{m_{\sigma(n)} m_{\tau(n)}}$$

(∵ For each $\sigma \in S(n)$, we can find $\prod_{i=1}^{\infty} n_i!$ τ's satisfying $m_{\sigma(i)} = m_{\tau(i)}$ $\forall i \in \mathbb{N}$ ∴)

$$= n! \prod_{i=1}^{\infty} n_i! .$$

→ $\left\{\frac{1}{\sqrt{n! \prod_{i=1}^{\infty} n_i!}} \sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}}, \quad m_1, \cdots, m_n \in \mathbb{N}\right\}$ is an ONB of $\mathcal{H}_{Sym}^{\otimes}$

Since $\frac{1}{\sqrt{n! \prod n_i!}} \sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}}$ can be identified by $\{m_1, \cdots, m_n\}$

we can denote it by $|n_1, n_2, \cdots \rangle$.

For simplicity, we also define $\mathcal{H}_{Sym}^{\otimes 0}$ as $\mathbb{C}$, with $\mathbb{C}$-basis $\{1\}$

→ we denote $1 := |0, 0, 0, \cdots \rangle$ (This is called the) vacuum state

Similarly,

$$a_k\left(\frac{1}{\sqrt{n!\prod_{i=1}^{\infty} n_i!}}\sum_{\sigma\in S_n} e_{m_{\sigma(1)}}\otimes\cdots\otimes e_{m_{\sigma(n)}}\right)$$

$$=\frac{\sqrt{n_k}}{\sqrt{(n-1)!(n_k-1)!\prod_{i\neq k} n_i!}}\sum_{\tau\in S_{n-1}} e_{m'_{\tau(1)}}\otimes\cdots\otimes e_{m'_{\tau(n-1)}}$$

$\rightarrow$ $$a_k\left(\sum_{\sigma\in S_n} e_{m_{\sigma(1)}}\otimes\cdots\otimes e_{m_{\sigma(n)}}\right)$$

$$=\sqrt{n}\, n_k \sum_{\tau\in S_{n-1}} e_{m'_{\tau(1)}}\otimes\cdots\otimes e_{m'_{\tau(n-1)}} \quad \text{— } (**)$$

where $\{m'_1,\dots,m'_{n-1}\}=\{m_1,\dots,m_n\}-\{k\}$

On the other hand,

$$\sum_{\sigma\in S_n}(e_{m_{\sigma(1)}}\otimes\cdots\otimes e_{m_{\sigma(n-1)}})\times e_k^*(e_{m_{\sigma(n)}})$$

$$=\sum_{\substack{\sigma\in S_n\\ m_{\sigma(n)}=k}} e_{m_{\sigma(1)}}\otimes\cdots\otimes e_{m_{\sigma(n-1)}}$$

$$=n_k\sum_{\tau\in S_{n-1}} e_{m'_{\tau(1)}}\otimes\cdots\otimes e_{m'_{\tau(n-1)}}$$

$\rightarrow$ A linear operator $a_k:\bigoplus_{n=0}^{\infty}\mathcal{H}^{\otimes n}\to\bigoplus_{n=0}^{\infty}\mathcal{H}^{\otimes n}$ s.t.

$$a_k(\psi_1\otimes\cdots\otimes\psi_n)=\sqrt{n}\,(\psi_1\otimes\cdots\otimes\psi_{n-1})\times e_k^*(\psi_n)$$

satisfies $(**)$

$\rightarrow$ $\{e_1,e_2,\dots\}$ corresponds to $\{a_1,a_2,\dots\}$ by this way

$\rightarrow$ we extend $A(f)(\psi_1\otimes\cdots\otimes\psi_n)=\sqrt{n}\,(\psi_1\otimes\cdots\otimes\psi_{n-1})\times\langle f,\psi_n\rangle$

(since $\psi_n, f\in\mathcal{H}$, $f^*(\psi_n):=\langle f,\psi_n\rangle$)

Lecture 7. Creation and annihilation operators.

For given ONB $\{e_1, e_2, \cdots\}$ of $\mathcal{H}$, the associated

creation : $a_k^+ |n_1, n_2, \cdots\rangle = \sqrt{n_k+1}\, |n_1, \cdots n_{k-1}, n_k+1, n_{k+1}, \cdots\rangle$.

i.e. $a_k^+ : \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \to \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}$ satisfy

$$a_k^+ \left(\frac{1}{\sqrt{n! \prod_{i=1}^{\infty} n_i!}} \sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}} \right)$$

$$= \frac{1}{\sqrt{(n+1)! \prod_{i=1}^{\infty} n_i!}} \sum_{\tau \in S_{n+1}} e_{m'_{\tau(1)}} \otimes \cdots \otimes e_{m'_{\tau(n+1)}} \quad \text{— } (*)$$

where $m_1' = m_1, \cdots, m_n' = m_n, \ m_{n+1}' = k$.

$\rightarrow$ $a_k^+ \left(\sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}} \right) = \frac{1}{\sqrt{n+1}} \sum_{\tau \in S_{n+1}} e_{m'_{\tau(1)}} \otimes \cdots \otimes e_{m'_{\tau(n+1)}}$

Note that for each $i = 1, \cdots, n+1$,

$\{\tau \in S_{n+1} : \tau(i) = n+1\}$ is isomorphic to S_n.

$\rightarrow$ A linear operator $a_k^+ : \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \to \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}$ s.t.

$$a_k^+ (\psi_1 \otimes \cdots \otimes \psi_n) = \frac{1}{\sqrt{n+1}} \Big(\psi_1 \otimes \cdots \otimes \psi_n \otimes e_k + \psi_1 \otimes \cdots \otimes \psi_{n-1} \otimes e_k \otimes \psi_n + \cdots + e_k \otimes \psi_1 \otimes \cdots \otimes \psi_n \Big).$$

satisfy $(*)$. So $\{e_k\}_{k \geq 1}$ corresponds to $\{a_k^+\}_{k \geq 1}$ by this.

$\rightarrow$ $A^+(f)(\psi_1 \otimes \cdots \otimes \psi_n) := \frac{1}{\sqrt{n+1}} \Big(\psi_1 \otimes \cdots \otimes \psi_n \otimes f + \cdots + f \otimes \psi_1 \otimes \cdots \otimes \psi_n \Big)$

We extend

If $\mathcal{H} = L^2(\mathbb{R})$, then

$$(A(f)\varphi)(x_1,\dots,x_{n-1}) = \sqrt{n}\int_{-\infty}^{\infty} \overline{f(y)}\ \varphi(x_1,\dots,x_{n-1},y)\,dy \qquad \forall \varphi \in L^2(\mathbb{R}^n)_{sym}$$

$$(A^+(f)\varphi)(x_1,\dots,x_{n+1}) = \frac{1}{\sqrt{n+1}}\sum_{j=1}^{n+1} f(x_j)\ \varphi(x_1,\dots,x_{j-1},x_{j+1},\dots,x_{n+1}) \qquad \forall \varphi \in L^2(\mathbb{R}^n)_{sym}$$

$a_k = A(e_k)$ $\quad a_k^+ = A^+(e_k)$

$\rightarrow$ $a(x) = A(\delta_x)$; $\quad a^+(x) = A^+(\delta_x)$
Heuristically

$$(a(x)\varphi)(x_1,\dots,x_{n-1}) = \sqrt{n}\,\varphi(x_1,\dots,x_{n-1},x)$$

$$(a^+(x)\varphi)(x_1,\dots,x_{n+1}) = \frac{1}{\sqrt{n+1}}\sum_{j=1}^{n+1} \delta(x-x_j)\,\varphi(x_1,\dots,\hat{x}_j,\dots,x_{n+1})$$

$\rightarrow$ $A(f) = \int dx\ \overline{f(x)}\ a(x)$, $\quad A^+(f) = \int dx\ f(x)\,a^+(x)$

$a_k a_l^+ - a_l^+ a_k = 0$ if $k \neq l$, since $|n_1, n_2, \dots\rangle \rightarrow \sqrt{n_l+1}\,\sqrt{n_k}\,|\cdots n_k - 1 \dots n_l + 1\rangle$

$\qquad\qquad\qquad\quad\ \ \mathbb{1}$ if $k = l$, since $a_k a_k^+ = (n_k+1)\mathbb{1}$
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad a_k^+ a_k = n_k \mathbb{1}$

On the other hand, since

$$[A(\textstyle\sum \alpha_k e_k),\ A^+(\sum \beta_k e_k)]$$
$$= \left[\sum_{k=1}^{\infty} \alpha_k^* a_k,\ \sum_{l=1}^{\infty} \beta_l a_l^+\right]$$
$$= \sum_{k,l} \alpha_k^* \beta_l\ [a_k, a_l^+]$$
$$= \sum_{k,l} \alpha_k^* \beta_l\ \delta_{k,l}\ \mathbb{1} = (\textstyle\sum \alpha_k e_k,\ \sum \beta_k e_k)\mathbb{1}$$

and $[A(f), A^+(g)] = \int dx\,dy\ \overline{f(x)}\ g(y)\ [a(x), a^+(y)]$,

we have $[a(x), a^+(y)] = \delta(x-y)\mathbb{1}$

Similarly, since $[a_k, a_k] = [a_k^+, a_k^+] = 0$, we have

$$[a(x), a(y)] = [a^+(x), a^+(y)] = 0 \qquad \forall x, y \in \mathbb{R}$$

Lecture 8 Time evolution on Fock space

If A is a linear operator (self-adj) on $L^2(\mathbb{R})$, $\varphi \in L^2(\mathbb{R}^n)_{sym}$, then

$$\left(\int dx\, a^+(x) A\, a(x)\right)(\varphi)(x_1, \dots, x_n)$$

$$(a(x)\varphi)(x_1, \dots, x_{n-1}) = \sqrt{n}\, \varphi(x_1, \dots, x_{n-1}, x)$$

$$(A_x a(x) \varphi)(x_1, \dots, x_{n-1}) = A_x\left(\sqrt{n}\, \varphi(x_1, \dots, x_{n-1}, x)\right)$$

$$(a^+(x) A_x a(x) \varphi)(x_1, \dots, x_n) = \sum_{j=1}^{n} \delta(x - x_j)\, A_x\left(\varphi(x_1, \dots, \hat{x}_j, \dots, x_n, x)\right)$$

$$= \sum_{j=1}^{n} A_{x_j}\left(\varphi(x_1, \dots, x_n)\right)$$

If A : multiplication operator f, then

$$\left(\int dx\, a^+(x) A\, a(x)\right)\varphi = \int dx\, f(x)\, a^+(x) a(x) \varphi = \sum_{j=1}^{n} A_{x_j} \varphi(x_1, \dots, x_n)$$
$$= \left(\sum_{j=1}^{n} f(x_j)\right) \varphi(x_1, \dots, x_n)$$

If A: free evolution Hamiltonian $H = -\frac{1}{2m}\frac{d^2}{dx^2}$

$$\to \left(\int_{\mathbb{R}} dx\, a^+(x) A\, a(x)\right)\varphi = \sum_{j=1}^{n} A_{x_j} \varphi(x_1, \dots, x_n)$$
$$= -\frac{1}{2m} \sum_{j=1}^{n} \frac{\partial^2}{\partial x_j^2} \varphi(x_1, \dots, x_n) = -\frac{1}{2m} \Delta \varphi$$

If $\varphi = \sum_{\sigma \in S_n} \varphi_1 \otimes \cdots \otimes \varphi_n$, then

$$\int dx\, a^+(x) A\, a(x)\, \varphi = \sum_{j=1}^{n} A_{x_j}\left(\varphi(x_1, \dots, x_n)\right) = \sum_{j=1}^{n} \varphi_1(x_1) \cdots \varphi_{j-1}(x_{j-1})\, (A\varphi_j)(x_j) \cdots \varphi_n(x_n)$$

This is exactly how Hamiltonian for tensor product works (Lecture 5.4)

I think there are nothing new in 8.4 and 8.5. So let's finish today's seminar.